Figure

Figure 3.1.1: Research Design Framework:

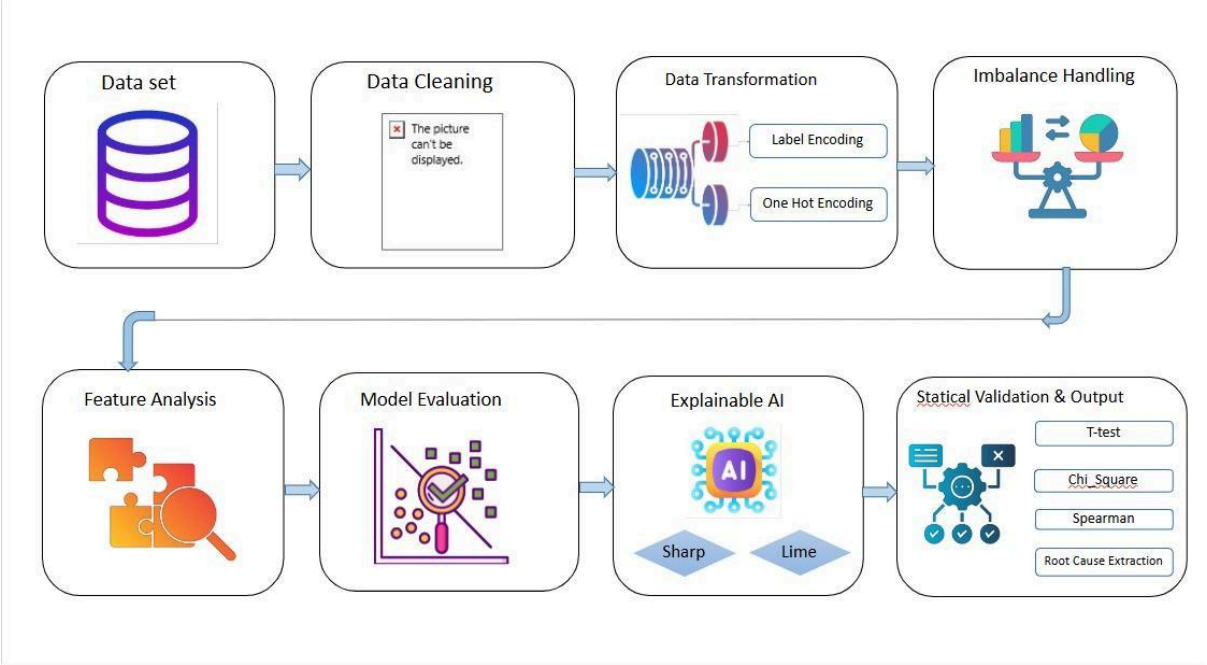


Figure 4.1.1: Age Distribution of the Respondents:

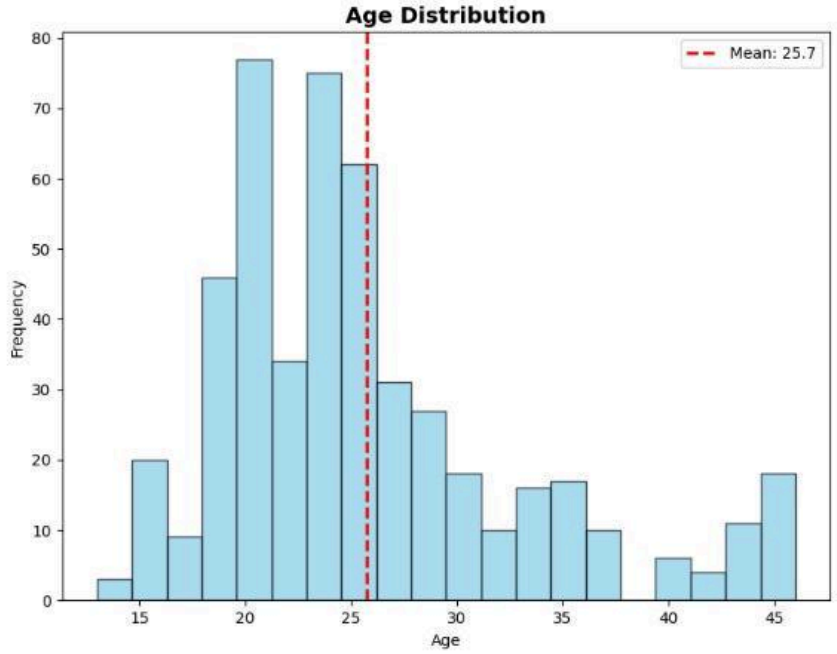


Figure 4.1.2: Gender Distribution of the Respondents:

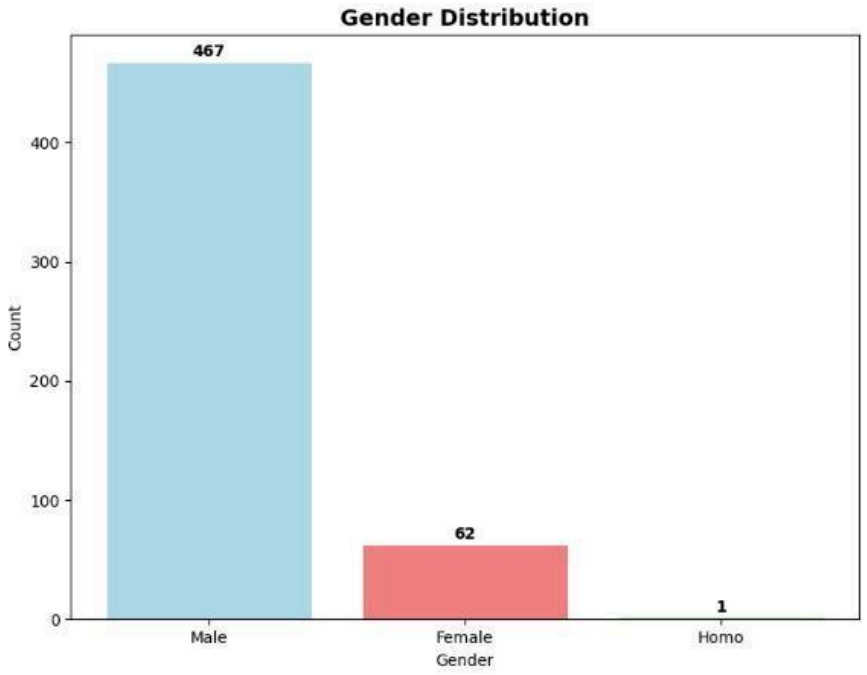


Figure 4.1.3: Location Distribution of the Respondents:

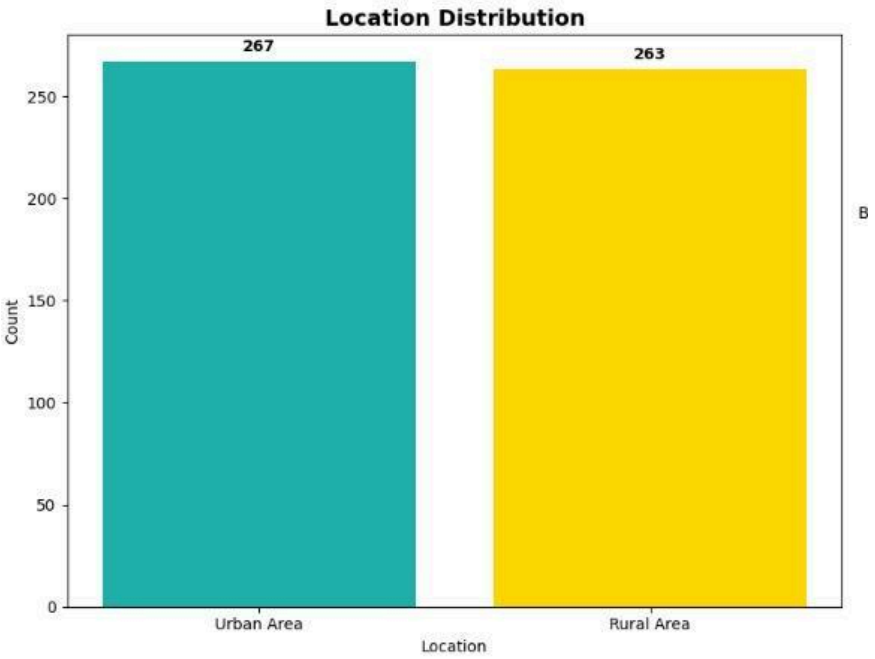


Figure 4.1.4: Distribution of Employment Types:

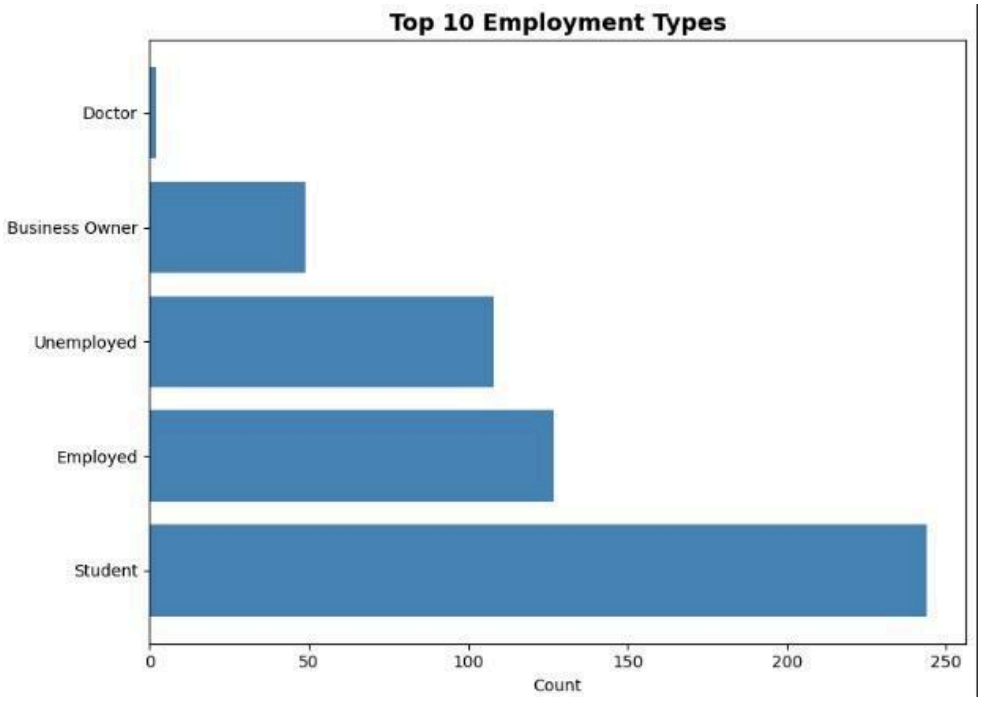


Figure 4.1.5: Drug Use Distribution:

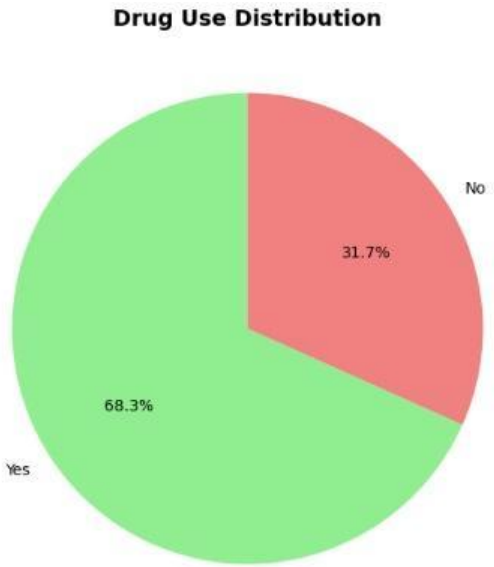


Figure 4.2.1: Correlation Heatmap:

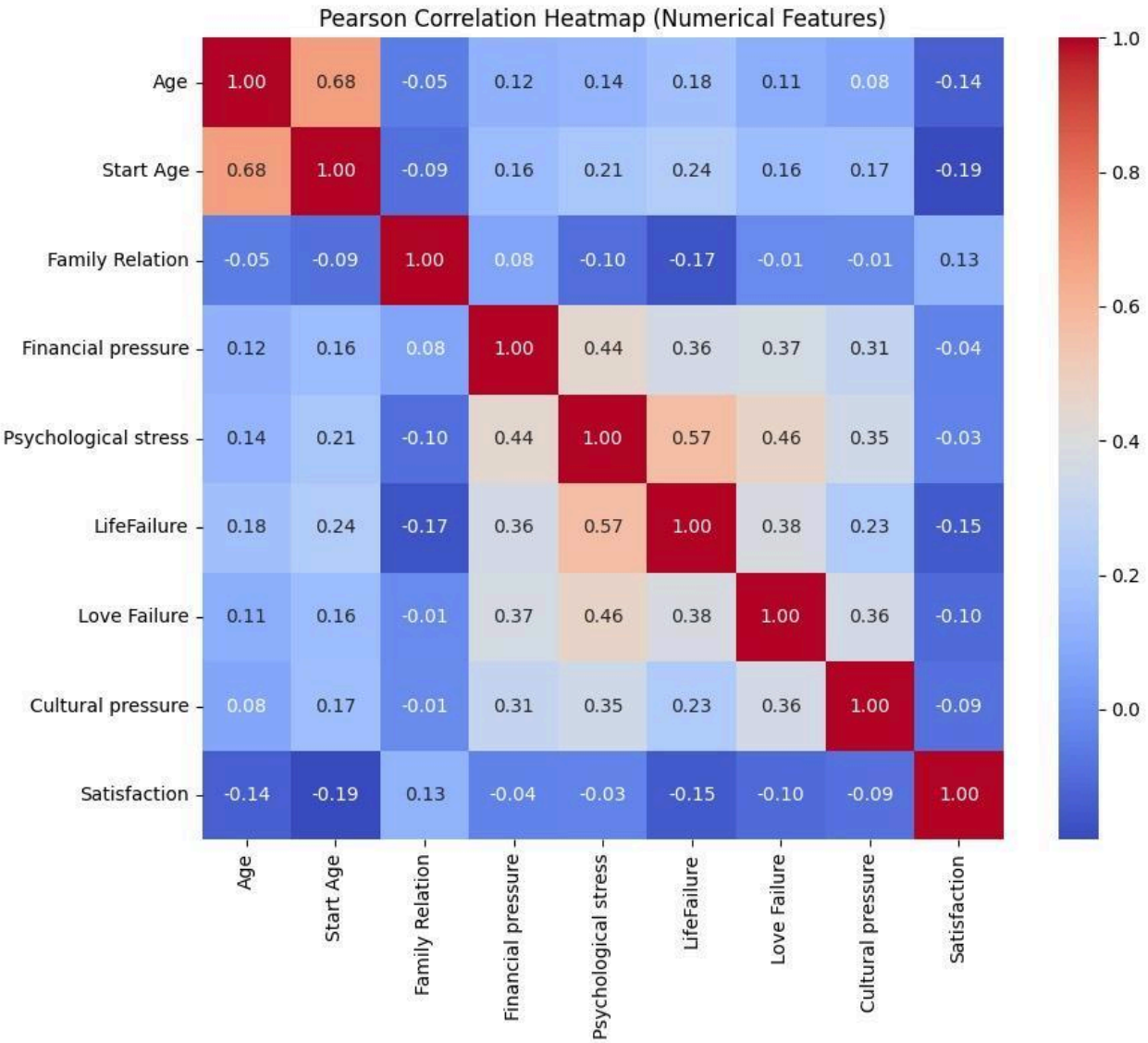


Figure 4.2.2 : Mutual Information Scores :

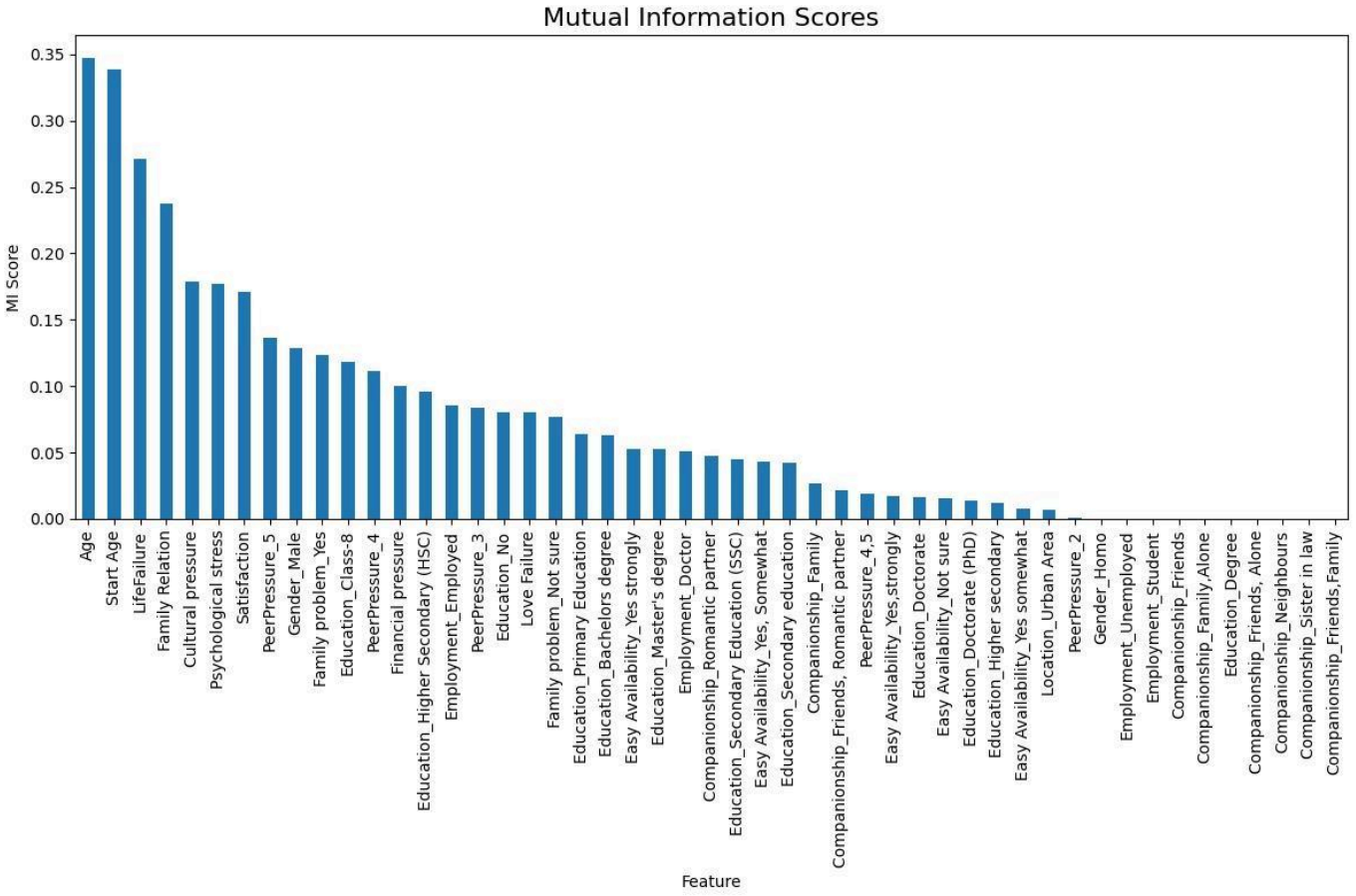


Figure 4.2.3 : Performance Comparison bar diagram:

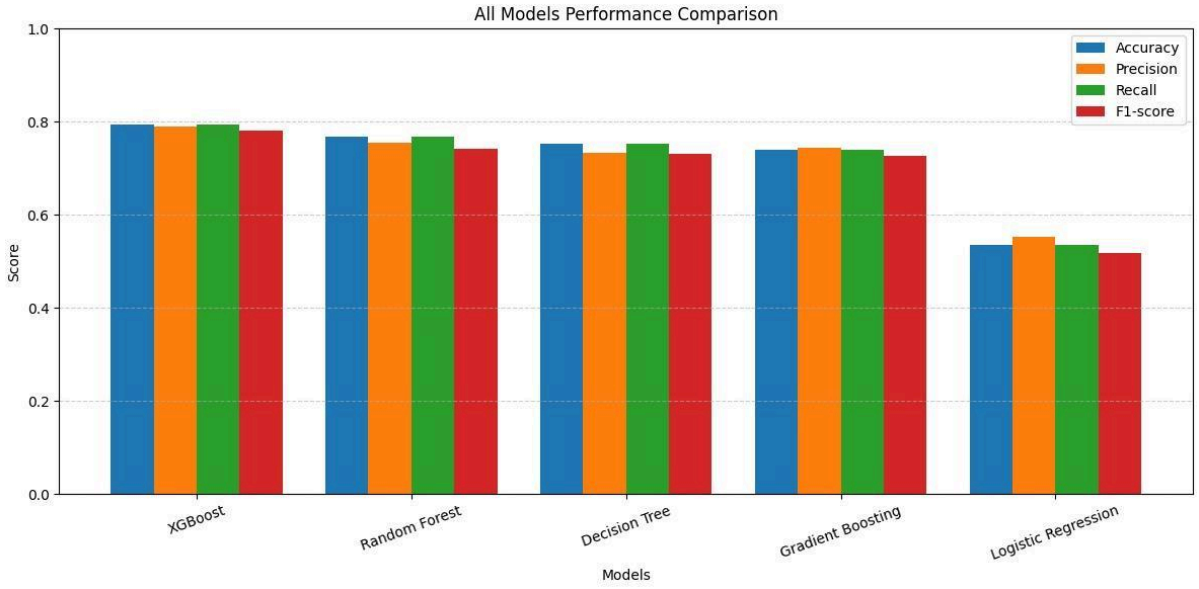


Figure 4.2.4: Confusion Matrix Analysis (XGBoost):

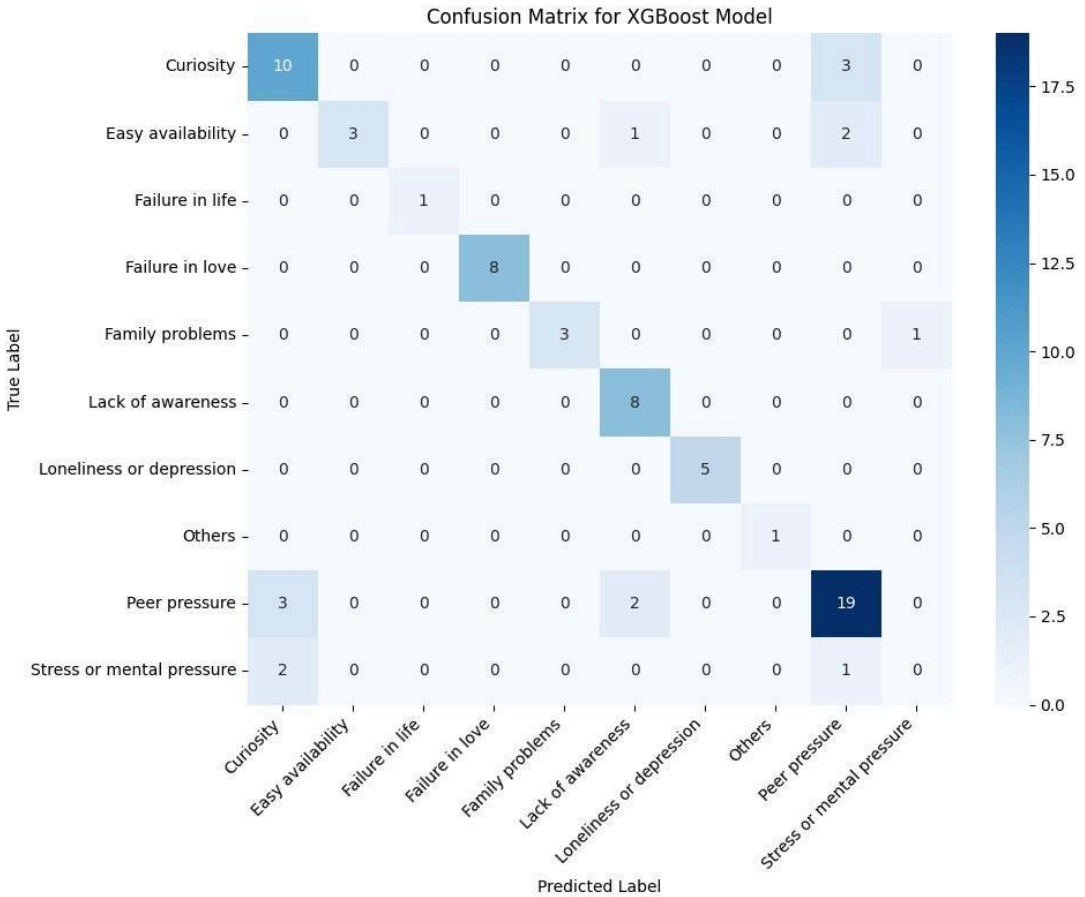


Figure 4.2.5 All models ROC Curve:

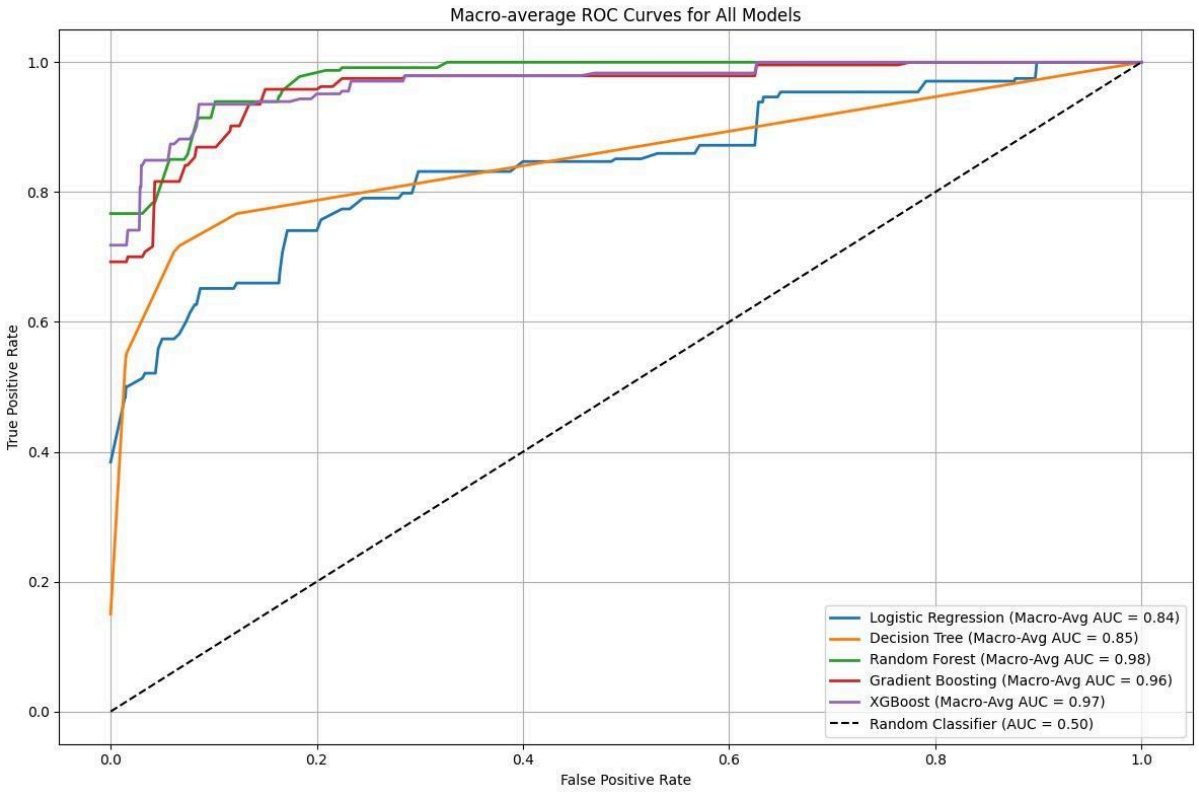


Figure 4.3.4: SHAP Global Feature Importance:

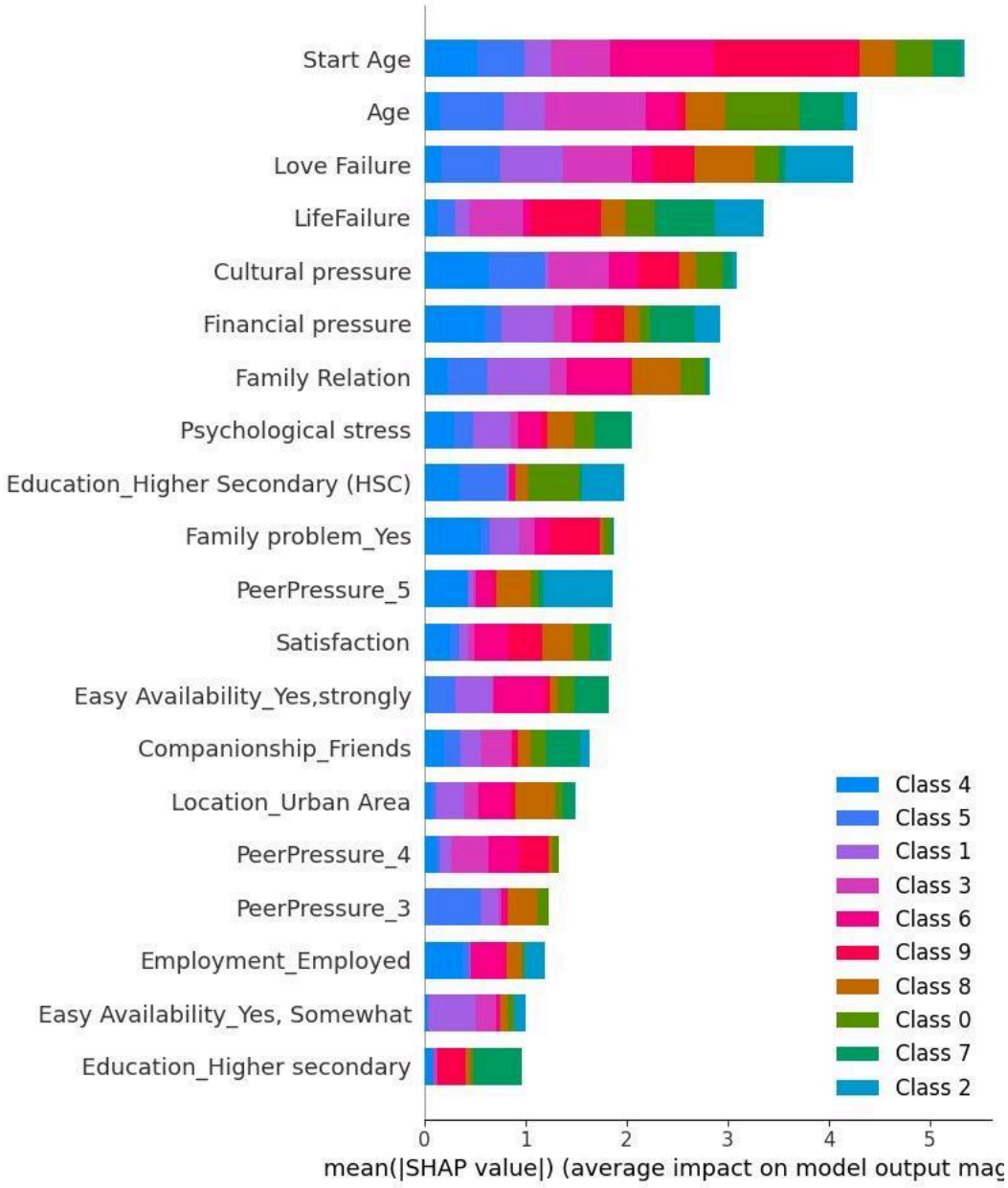


Figure 4.3.5: SHAP Dependence Plot:

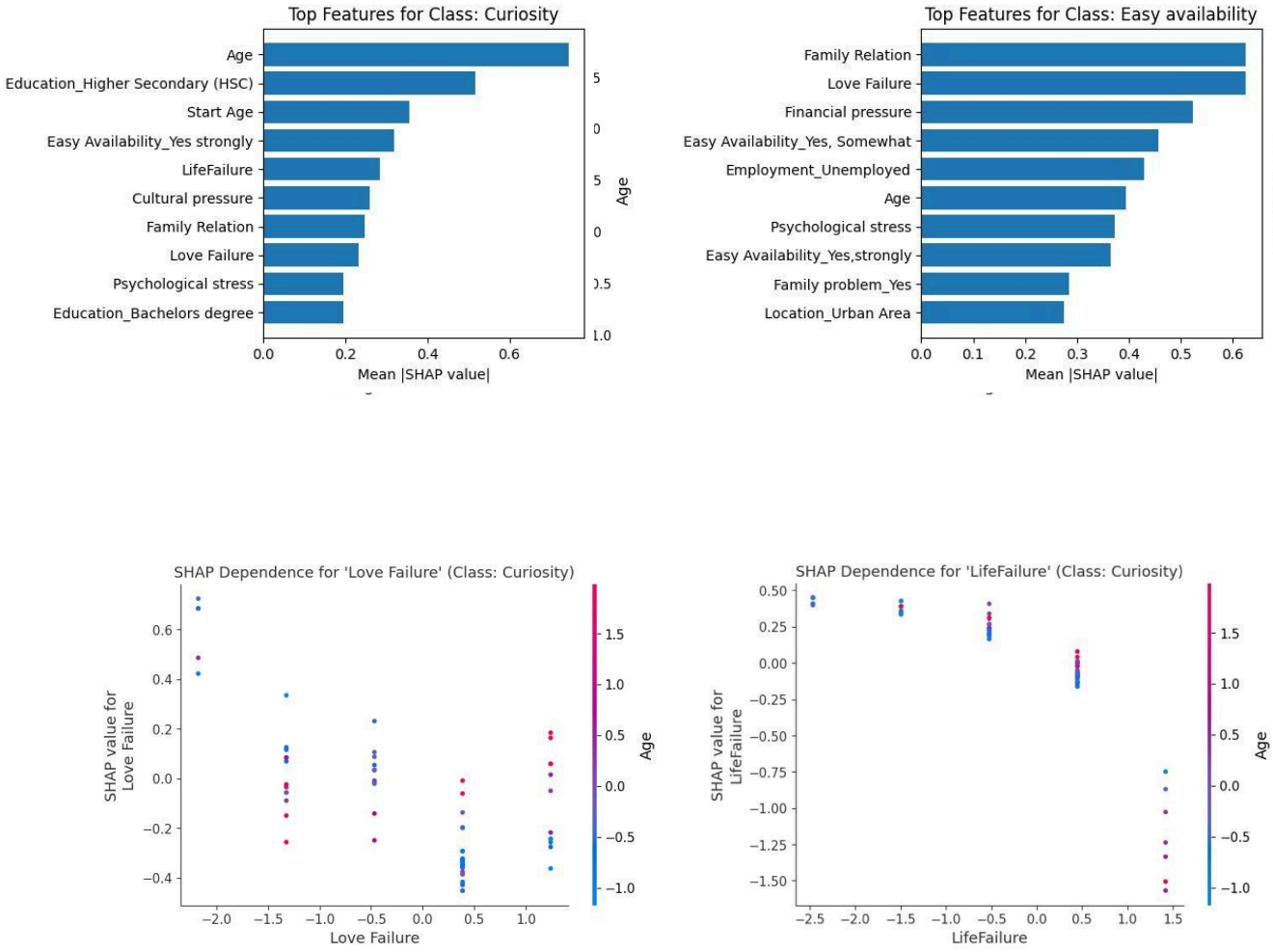


Figure 4.3.6: Class-wise SHAP Feature Importance Analysis:



Figure 4.3.7: LIME Local Explanation:

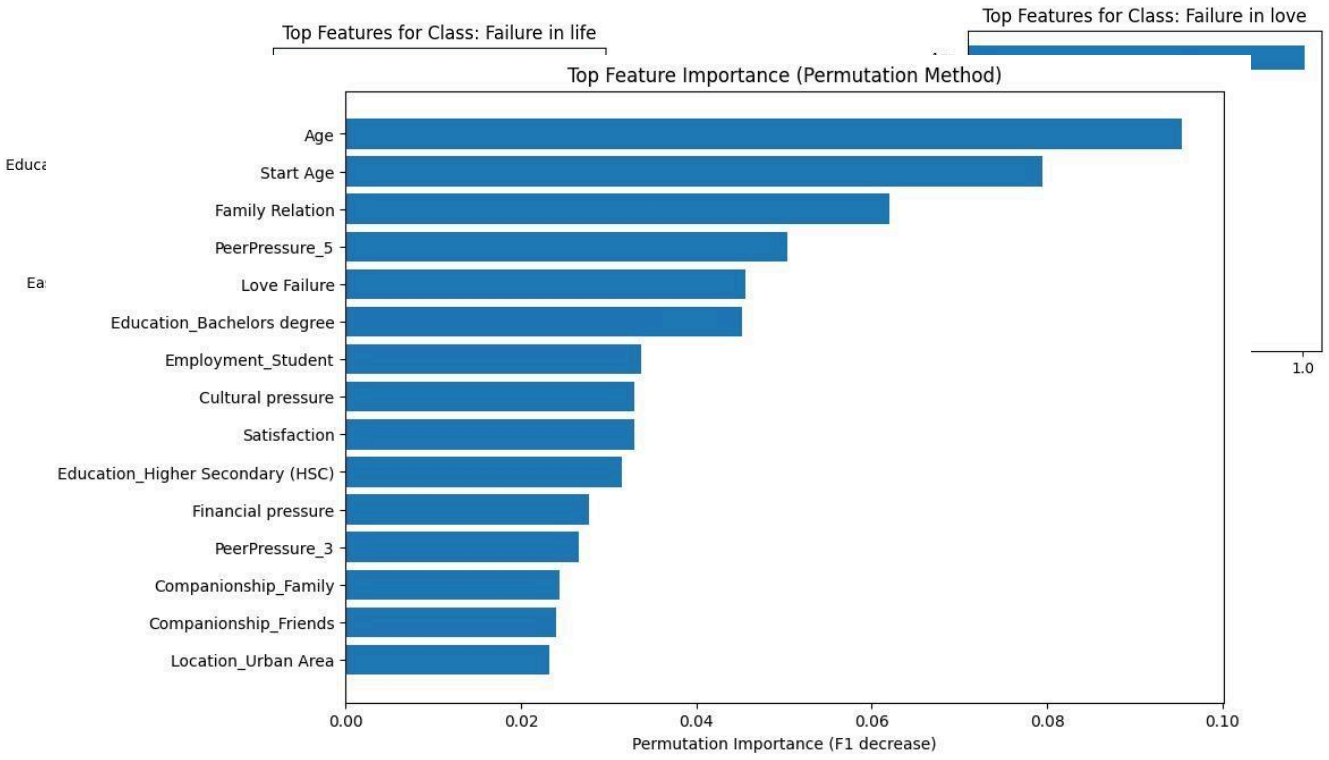
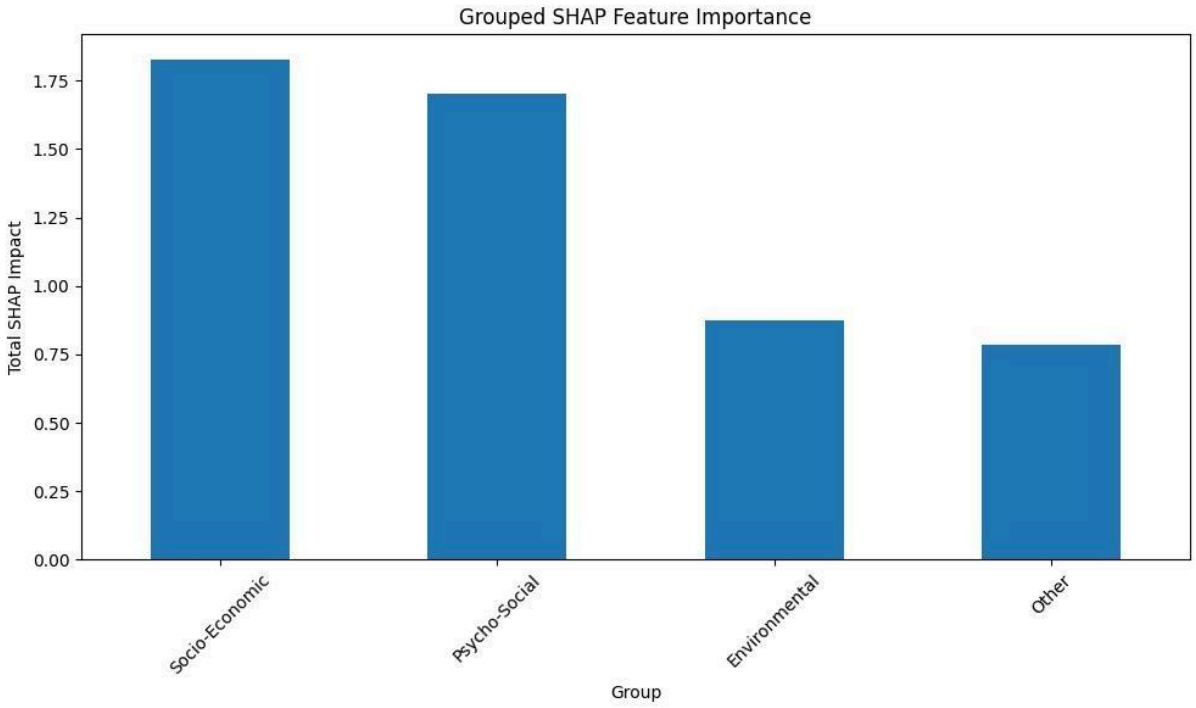


Figure 4.3.8: Permutation Feature Importance:

Figure 4.3.9: Grouped SHAP Feature Importance for Drug Addiction Prediction:



Table

Table 2.1: Summary of Literature Reviewed

Author(s)	Year	Title	Methodology	Key Findings
Shazzad et al.[9]	2013	Causes of Drug Addiction in Bangladesh	Survey	Identified socio-cultural and family-related causes; no AI used.
Khanam et al.[5]	2024	Drug Use Prediction with XAI	ML + LIME	Explained influence of stress and impulsiveness on addiction risk.
Arif et al.[7]	2021	Addiction Risk Prediction Using ML	ML models (97.9% accuracy)	High accuracy; lacks interpretability.
Bennett et al.[12]	2013	Lifestyle Factors in Student Drug Use	Logistic Regression	Linked nightlife and independence with drug use.
Tito et al[4]	2024	Early Detection of Drug Addiction Risk	ML models (95% accuracy)	Effective early detection; no explainability.
Zhang et al. [13]	2019	Drug and Alcohol Risk Prediction	Logistic Regression	Achieved 97.91% accuracy; strong predictors of substance risk identified.
Zhu et al. [10]	2017	Smoking Behavior Prediction	XGBoost	Achieved 84.11% accuracy; identified behavior-related smoking factors.
Zhant et al. [8]	2018	Tobacco Disease Detection	ELM	Achieved 98.3% accuracy in tobacco-related disease detection.
Mutamed et al. [14]	2025	Investigating youth drug addiction causes	Survey	Identified Addiction arises from multifactor influences.
Lee et al. [6]	2019	Smoking Status Prediction	Logistic Regression	Achieved 83.44% accuracy.
Kinreich et al. [2]	2019	Alcohol Use Disorder Treatment Seeking	Logistic Regression	Identified factors influencing treatment seeking (NM1).
Kumari et al. [1]	2018	Alcohol Use Disorder Risk Prediction	NM1	Highlighted key risk indicators (NM1).
Abdul Ahad et al. [13]	2022	Youth drug addiction causes in Sylhet	Survey	Abdul Ahad et al. [21]

Table 2.3: Gap Analysis of Existing Studies and the Proposed Research:

Features	Shazzad et al.[9]	Khanam et al.[5]	Arif et al. [7]	Tito et al.[4]	Zhu et al. [10]	Proposed system
Uses real-world data (Bangladesh)	Yes	No	Yes	Yes	No	Yes
Machine Learning applied	No	Yes	Yes	Yes	Yes	Yes
Explainable AI (XAI) used	No	Yes(LIME only)	No	No	Rare	Yes (SHAP + LIME)
Identifies root causes of addiction	Partially	Yes	Partially	No	Limited	Yes
High prediction accuracy	No	Yes	Yes	Yes	Yes	Yes
Transparent explanations	No	Limited	No	No	No	Yes
Focus on Bangladesh context	Yes	No	Yes	Yes	No	Yes
Multi-factor dataset (social + psychological + economic)	Limited	Limited	Moderate	Limited	Varies	Yes
Suitable for policy and rehabilitation decisions	No	Partially	No	No	No	Yes

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Data collection phase																			
Preprocess all the data																			
Feature Analysis																			
Model training																			
XAI																			
Statical Validation																			

Table 3.3 Task Allocation

Table 4.1: Experimental Setup:

Category	Tools/Details Used
Operating System	Windows 10 (64-bit)
Development Platform	Google Colab
Programming Language	Python 3.x
ML Frameworks	Scikit-learn, XGBoost
XAI Libraries	SHAP, LIME
Supporting Libraries	NumPy, Pandas, Matplotlib, Seaborn
Statistical Tools	SciPy

Runtime Environment	CPU (Intel Core i5, 8GB RAM)
Data Storage	Local Drive + Google Drive

Table 4.2.4 Performance Summary of All Machine Learning Models:

Model	Accuracy	Precision	Recall	F1-score
XGBoost	0.7945	0.7908	0.7945	0.7817
Random Forest	0.7671	0.754	0.7671	0.7427
Decision Tree	0.7534	0.7331	0.7534	0.7307
Gradient Boosting	0.7397	0.7441	0.7397	0.7262
Logistic Regression	0.5342	0.5528	0.5342	0.5185

Table 4.2.7: Cross-Validation Performance :

Model	Accuracy (Mean ± Std)	Precision (Mean ± Std)	Recall (Mean ± Std)	F1-score (Mean ± Std)
Random Forest	0.8094 ± 0.0165	0.8204 ± 0.0330	0.8094 ± 0.0165	0.7940 ± 0.0192
XGBoost	0.7791 ± 0.0317	0.7726 ± 0.0486	0.7791 ± 0.0317	0.7652 ± 0.0338
Gradient Boosting	0.7708 ± 0.0180	0.7699 ± 0.0348	0.7708 ± 0.0180	0.7596 ± 0.0189
Decision Tree	0.7597 ± 0.0416	0.7807 ± 0.0519	0.7597 ± 0.0416	0.7548 ± 0.0440
Logistic Regression	0.5193 ± 0.0352	0.5130 ± 0.0360	0.5193 ± 0.0352	0.4944 ± 0.0387

Table 4.2.8: Model Stability Test:

Random State	Accuracy	Macro F1-score
10	0.7945	0.7891
20	0.7397	0.7318
30	0.7123	0.7108
40	0.7945	0.7841
50	0.7671	0.7676

Table 4.3.1: Dataset Description:

Item	Description
Total samples (after filtering)	362
Total features (before encoding)	17
Encoded features	49
Target variable	Reason for drug addiction
Number of target classes	10
Train–test split	80% training, 20% testing

Table 4.3.2: Distribution of Drug Addiction Reasons:

Reason	Frequency
Peer pressure	117
Curiosity	66
Failure in love	37
Lack of awareness	37
Easy availability	30
Loneliness or depression	23
Family problems	22
Stress or mental pressure	17
Failure in life	7
Others	6

Table 4.4.1: ANOVA Test Results (Numerical Features vs. Reason) :

Feature	F-Statistic	p-value	Significant ($p < 0.05$)
Age	2.6991	0.004763	Yes
Start Age	4.0159	0.000066	Yes
Family Relation	6.306	0	Yes
Financial Pressure	2.1842	0.022627	Yes
Psychological Stress	3.3893	0.000525	Yes
Life Failure	5.5939	0	Yes
Love Failure	4.7486	0.000006	Yes
Cultural Pressure	3.5529	0.000307	Yes
Satisfaction	1.1591	0.320468	No

Table 4.4.2 : Chi-Square Test Results

Feature	Chi-square	p-value	Significant ($p < 0.05$)
Gender	12.9892	0.792213	No
Employment	54.2948	0.025758	Yes
Education	444.5624	0	Yes
Location	15.4711	0.078783	No
Companionship	172.9729	0	Yes
Family Problem	63.8487	0	Yes
Peer Pressure	182.763	0	Yes
Easy Availability	100.0914	0.000005	Yes
Category	1424	0	Yes

Table 4.4.3: Pearson Correlation Results:

Feature	Pearson r	p-value	Significant
Age	–0.0913	0.082948	No
Start Age	–0.0170	0.747023	No
Family Relation	0.0874	0.096717	No
Financial Pressure	0.0642	0.223129	No
Psychological Stress	–0.0803	0.127038	No
Life Failure	0.0326	0.536232	No
Love Failure	–0.0873	0.097323	No
Cultural Pressure	–0.0509	0.333729	No
Satisfaction	–0.0359	0.495359	No

Feature	Spearman ρ	p-value	Significant
Age	−0.0918	0.081018	No
Start Age	0.0465	0.37729	No
Family Relation	0.0533	0.311462	No
Financial Pressure	0.0636	0.227024	No
Psychological Stress	−0.0767	0.145385	No
Life Failure	0.0692	0.188835	No
Love Failure	−0.0727	0.167633	No
Cultural Pressure	−0.0631	0.231228	No
Satisfaction	−0.0435	0.408814	No

Table 4.4.4: Spearman Correlation Results:

Table 4.4.5: SHAP-Based Identification of Root Causes:

Rank	Feature	Mean SHAP Impact	Impact Direction
1	Start Age	–0.320952	Decreases Risk
2	Age	–0.213168	Decreases Risk
3	Life Failure	–0.164491	Decreases Risk
4	Love Failure	–0.145759	Decreases Risk
5	Cultural Pressure	–0.126733	Decreases Risk
6	Family Relation	–0.094716	Decreases Risk
7	Psychological Stress	–0.093863	Decreases Risk
8	Peer Pressure (Level 5)	–0.078441	Decreases Risk
9	Financial Pressure	–0.073146	Decreases Risk

Table 5.3. Expense Category and Cost Summary:

Serial	Expense Category	Description	Estimated Cost (TK)
1	Data Collection	Travel to rehabilitation centers, coordination with DNC-authorized staff, phone calls, field communication, and follow-up interviews.	10000
2	Printing Costs	Printing research forms, questionnaires, consent papers, awareness posters, and final report copies.	3000
3	T-Shirt Costs	T-shirts used for awareness campaigns and team identification during field activities.	3000
5	Additional Field Interviews	Expanding dataset with more rehabilitation centers.	5000
		Total =	21000

Table 5.1: Mapping with Complex Engineering Problem:

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
✓	✗	✓	✓	✓	✓	✓

Table 5.2: Mapping with knowledge Profile:

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
✗	✓	✓	✓	✓	✓	✗	✓

Table 5.3: Mapping with Complex Engineering Activities:

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓	✓	✓	✓